

Deutsch's Algorithm

→ Deutsch's Problem

→ Parity problem for $f: \Sigma \rightarrow \Sigma$

Input → $f: \Sigma \rightarrow \Sigma$

Output → 0 if f is constant,
1 if f is balanced

$f_1, f_4 \rightarrow$ Constant
 $f_2, f_3 \rightarrow$ Balanced

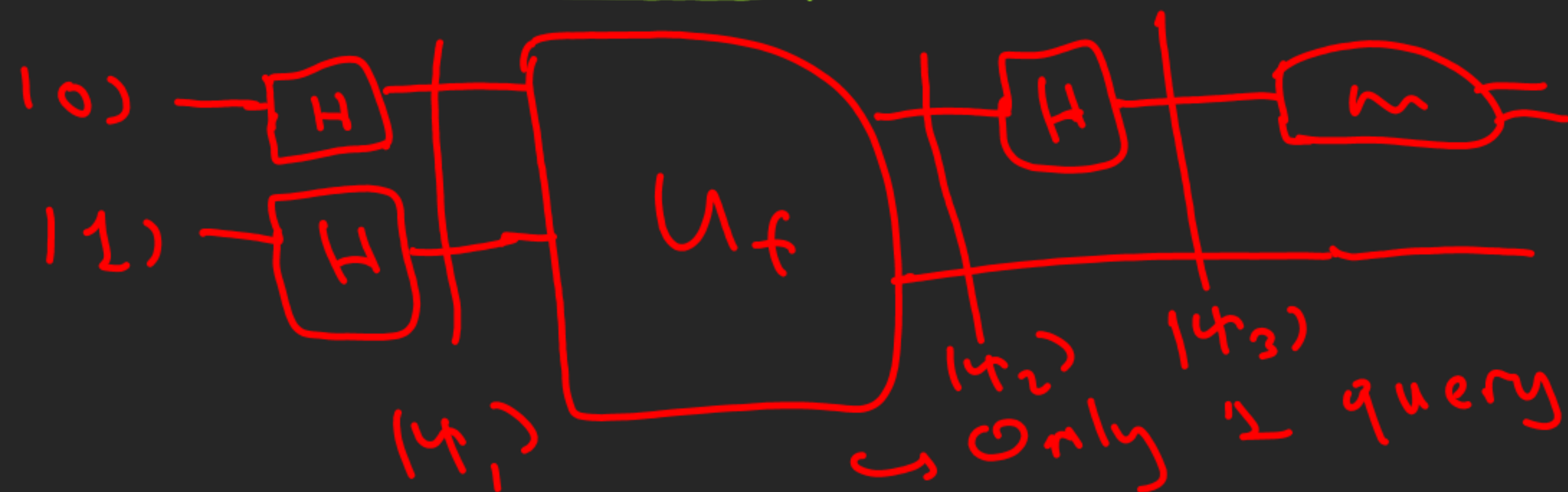
a	$f_1(a)$
0	0
1	0

a	$f_2(a)$
0	0
1	1

a	$f_3(a)$
0	1
1	0

a	$f_4(a)$
0	2
1	2

Deutsch's Algorithm



$\begin{cases} 0 & \text{if } f \text{ is constant} \\ 1 & \text{if } f \text{ is balanced} \end{cases}$

Classically takes at least 2 queries

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$\begin{aligned}
 |\psi_2\rangle &= U_f(|\psi_1\rangle) = U_f\left(\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\right) \\
 &= U_f\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle + |10\rangle - |11\rangle)\right)
 \end{aligned}$$

Phase kickback

$$|b \oplus c\rangle = X^c |b\rangle$$

$$U_f(|a\rangle |b\rangle) = |a\rangle |b \oplus f(a)\rangle = |a\rangle (X^{f(a)} |b\rangle)$$

$$|b\rangle = |-\rangle \rightarrow U_f(|a\rangle |-\rangle) = |a\rangle (X^{f(a)} |-\rangle)$$

$$X|-\rangle = -|-\rangle$$

$$= (-1)^{f(a)} |a\rangle |-\rangle$$

$$U_f(|a\rangle |-\rangle) = (-1)^{f(a)} |a\rangle |-\rangle$$

